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ToDo Code:

```
def test_add_sheep():  @ Mike

# TODO: Prepare the new sheep data in a dictionary format.
new_sheep = {
    "id": 7,
    "name": "Suffern",
    "breed": "Suffolk",
    "sex": "ram"
}

# TODO: Send a POST request to the endpoint "/sheep" with the new sheep data.
# Arguments should be your endpoint and new sheep data.
response = client.post(url="/sheep", json=new_sheep)

# TODO: Assert that the response status code is 201 (Created)
assert response.status_code == 201
# TODO: Assert that the response JSON matches the new sheep data
assert response.json() == new_sheep
# TODO: Verify that the sheep was actually added to the database by retrieving the new sheep by ID.
# include as assert statement to see if the new sheep data can be retrieved.
get_response = client.get(f"/sheep/{new_sheep['id']}")
assert get_response.status_code == 200
assert get_response.json() == new_sheep
```

Test Case (2 Passed):

```
(.venv) PS C:\Users\mmal\IdeaProjects\sheep_compendium> pytest
===== test session starts =====
platform win32 -- Python 3.10.9, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\mmal\IdeaProjects\sheep_compendium
plugins: anyio-4.14.2
collected 2 items

tests/test_main.py .. [100%]

===== warnings summary =====
.venv\lib\site-packages\fastapi\testclient.py:1
C:\Users\mmal\IdeaProjects\sheep_compendium\.venv\lib\site-packages\fastapi\testclient.py:1: StarletteDeprecationWarning: Using 'httpx' with 'starlette.testclient' is deprecated; install 'httpx2' instead.
  from starlette.testclient import TestClient as TestClient # noqa
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 2 passed, 1 warning in 0.50s =====
(.venv) PS C:\Users\mmal\IdeaProjects\sheep_compendium>
```

SwaggerUI Post Response Body:

Response body

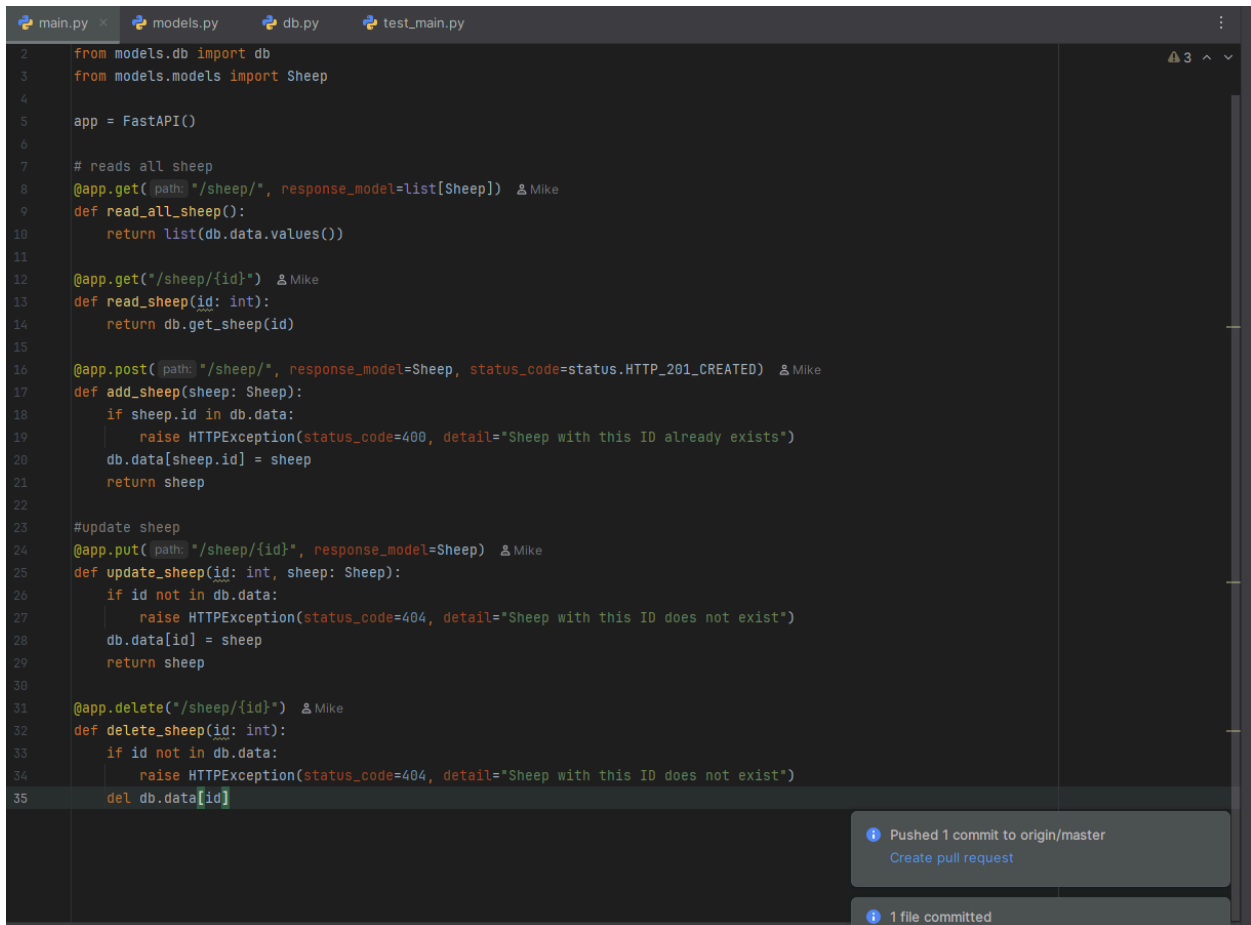
```
{
  "id": 7,
  "name": "Suffern",
  "breed": "Suffolk",
  "sex": "ram"
}
```

SwaggerUI Get Response Body:

Response body

```
{
  "id": 2,
  "name": "Blondie",
  "breed": "Polypay",
  "sex": "ram"
}
```

All CRUD operations coded:



```
1 from models.db import db
2 from models.models import Sheep
3
4
5 app = FastAPI()
6
7 # reads all sheep
8 @app.get(path="/sheep/", response_model=list[Sheep])  ⚡ Mike
9 def read_all_sheep():
10     return list(db.data.values())
11
12 @app.get("/sheep/{id}")  ⚡ Mike
13 def read_sheep(id: int):
14     return db.get_sheep(id)
15
16 @app.post(path="/sheep/", response_model=Sheep, status_code=status.HTTP_201_CREATED)  ⚡ Mike
17 def add_sheep(sheep: Sheep):
18     if sheep.id in db.data:
19         raise HTTPException(status_code=400, detail="Sheep with this ID already exists")
20     db.data[sheep.id] = sheep
21     return sheep
22
23 #update sheep
24 @app.put(path="/sheep/{id}", response_model=Sheep)  ⚡ Mike
25 def update_sheep(id: int, sheep: Sheep):
26     if id not in db.data:
27         raise HTTPException(status_code=404, detail="Sheep with this ID does not exist")
28     db.data[id] = sheep
29     return sheep
30
31 @app.delete("/sheep/{id}")  ⚡ Mike
32 def delete_sheep(id: int):
33     if id not in db.data:
34         raise HTTPException(status_code=404, detail="Sheep with this ID does not exist")
35     del db.data[id]
```

Pushed 1 commit to origin/master
Create pull request

1 file committed

- One reads all sheep (list of each sheep in db)
- We can read a sheep based on the specified ID.
- Add a sheep into the database.
- Update a sheep description.
- Delete a sheep description based on ID.

All Tests Screenshot:

```
.venv) PS C:\Users\mmal\IdeaProjects\sheep_compendium> pytest
===== test session starts =====
Platform win32 -- Python 3.10.9, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\mmal\IdeaProjects\sheep_compendium
plugins: anyio-4.14.2
collected 5 items

tests\test_main.py ..... [100%]

===== warnings summary =====
.venv\lib\site-packages\fastapi\tutorial.py:1
C:\Users\mmal\IdeaProjects\sheep_compendium\.venv\lib\site-packages\fastapi\tutorial.py:1: StarletteDeprecationWarning: Using 'httpx' with 'starlette.testclient' is deprecated; install 'httpx2' instead.
  from starlette.testclient import TestClient as TestClient # noqa

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 5 passed, 1 warning in 0.57s =====
.venv) PS C:\Users\mmal\IdeaProjects\sheep_compendium>
```

Test Code:

```
def test_read_all_sheep():
    """new *"""
    response = client.get("/sheep")
    sheep_list = response.json()

    assert response.status_code == 200
    assert len(sheep_list) >= 6
    assert {
        "id": 1,
        "name": "Spice",
        "breed": "Gotland",
        "sex": "ewe"
    } in sheep_list

def test_update_sheep():
    """new *"""
    updated_sheep = {
        "id": 2,
        "name": "Blondie",
        "breed": "Polypay",
        "sex": "ewe" #initially was 'ram'
    }

    response = client.put(url=f"/sheep/{updated_sheep['id']}", json=updated_sheep)

    assert response.status_code == 200
    assert response.json() == updated_sheep

def test_delete_sheep():
    """new *"""
    sheep_to_delete = {
        "id": 3,
        "name": "Deedee",
        "breed": "Jacobs Four Horns",
        "sex": "ram"
    }

    response = client.delete(f"/sheep/{sheep_to_delete['id']}")
    assert response.status_code == 204
    get_response = client.get(f"/sheep/{sheep_to_delete['id']}")
    assert get_response.status_code == 200
    assert get_response.json() is None
```

In `test_read_all_sheep`, I had it test to ensure that there was a list of sheep and that the sheep in `'id = 1'` matched what we had in the database.

In `test_update_sheep`, I had it test an updated value.

For `test_delete_sheep`, I had it make sure that the sheep in `id = 3` returned nothing as I would have deleted it from the database.